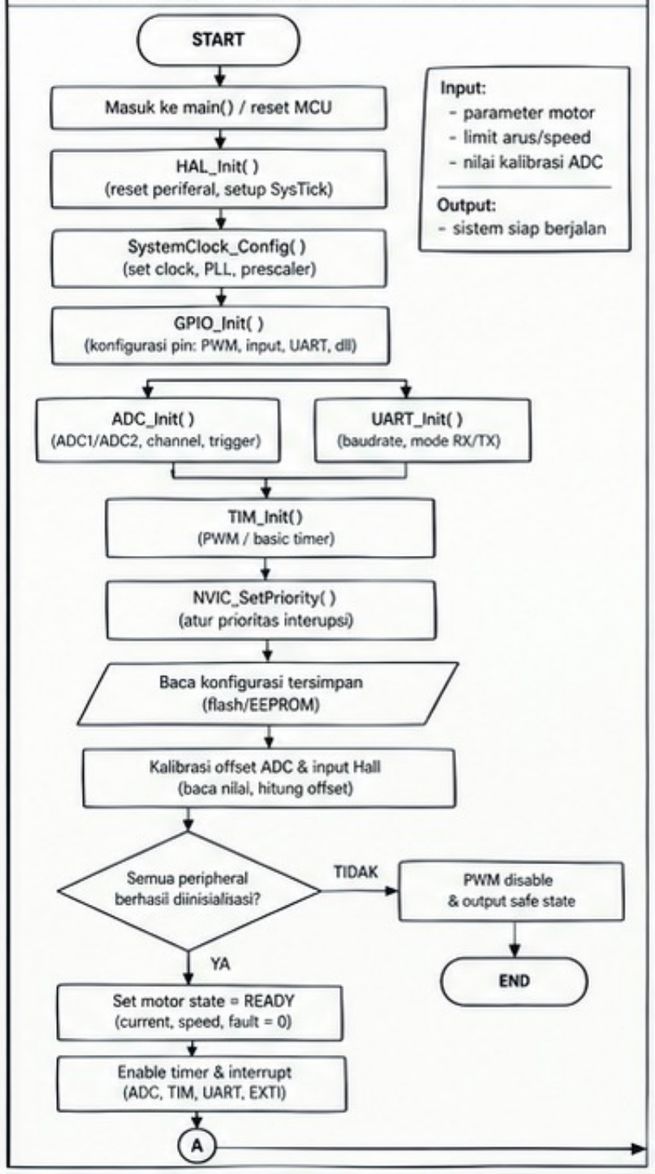
PRAKTIKUM SISTEM PENGENDALI ELEKTRIK — REVERSE ENGINEERING FIRMWARE BLDC/FOC

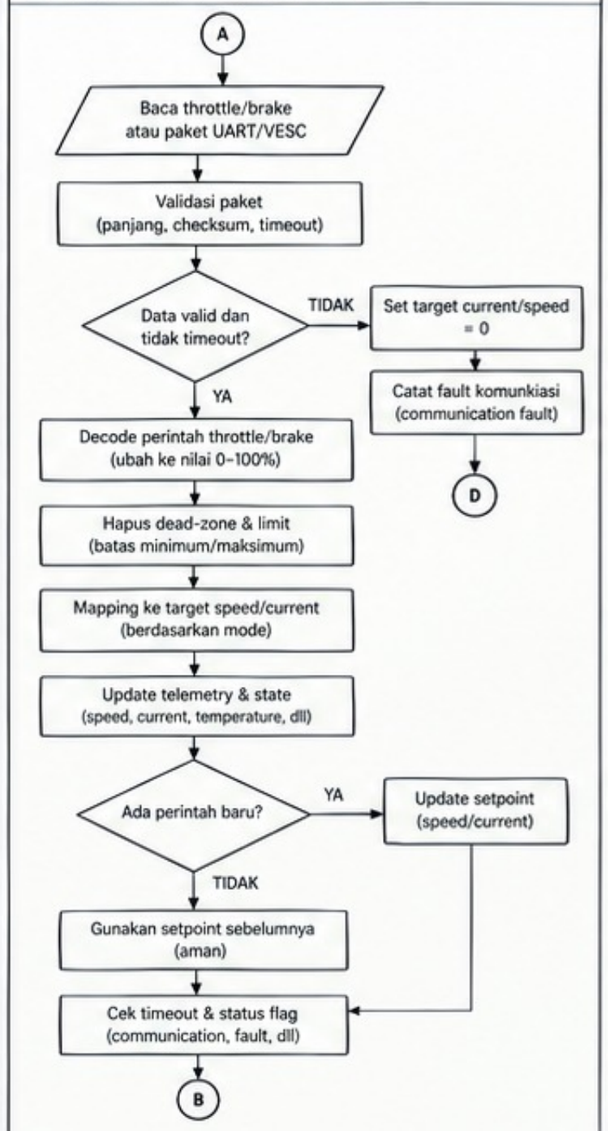
Purwo Daffarel | NIM 40040324620003 | Kelas Praktikum A

2. SmartESC_STM32_v2 — Repository 2

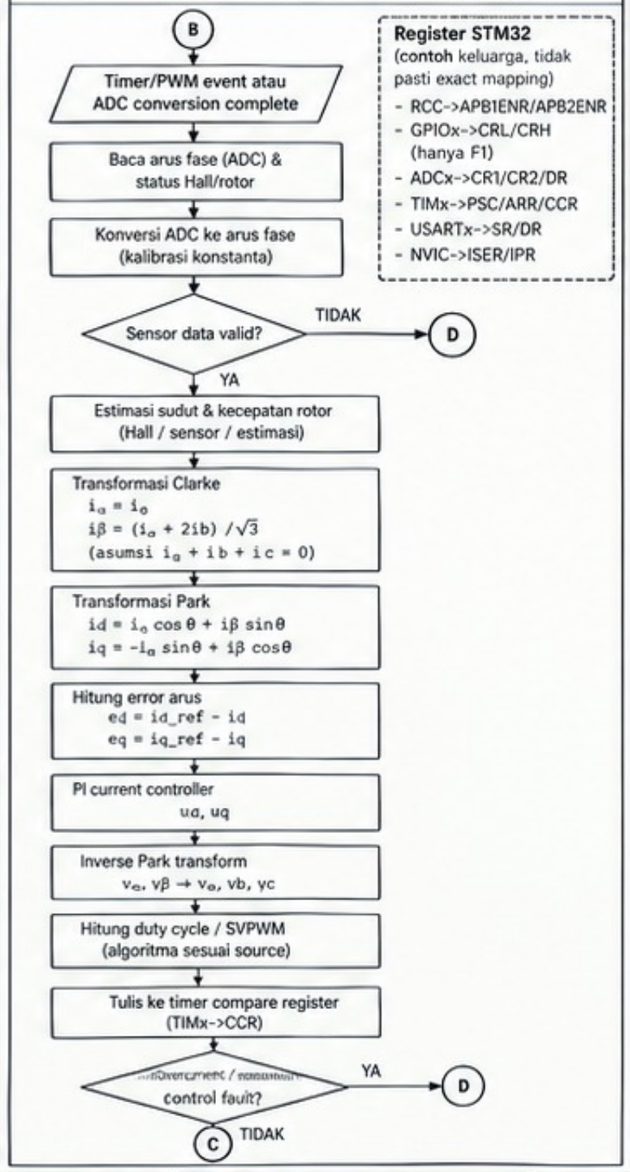
I. SYSTEM INITIALIZATION & SETUP ROUTINE



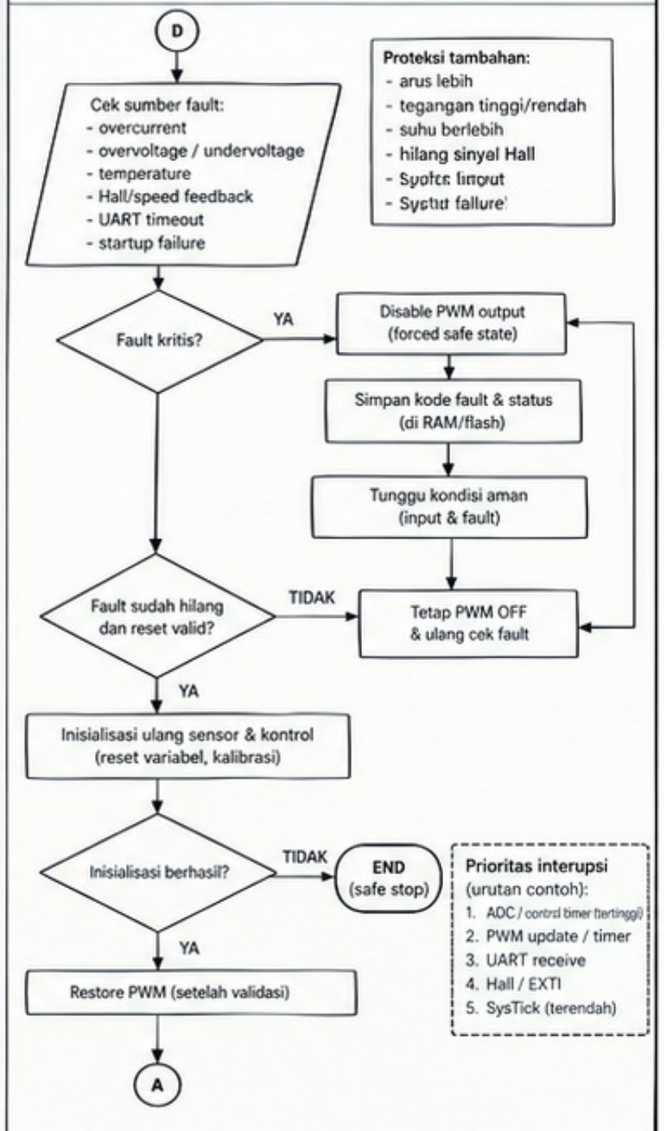
II. MAIN LOOP / TASK SCHEDULER



III. MOTOR CONTROL / HIGH-FREQUENCY ISR



IV. FAULT HANDLING & SAFETY ROUTINE



V. CONTOH PROGRAM (C/C++)

Fungsi utama:

```
int main(void) {
    HAL_Init();
    SystemClock_Config();
    MX_GPIO_Init();
    MX_ADCx_Init();
    MX_TIMx_Init();
    MX_USARTx_Init();
    while (1) {
        read_command();
        validate_command();
        update_setpoint();
        check_timeout();
    }
}
```

Contoh ISR & callback:

```
void HAL_ADC_ConvCpltCallback(ADC_HandleTypeDef* hadc) {
    read_adc_values();
    clarke_transform();
    park_transform();
    pi_controller();
    inverse_park();
    set_pwm_compare();
}
void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef* htim)
    motor_control_step();
}
void HAL_UART_RxCpltCallback(UART_HandleTypeDef* huart) {
    parse_uart_data() ;
}
void Error_Handler(void) {
    while (1) { /* error */ }
```

VI. PSEUDOCODE

Main Loop:

```
main() {
    HAL_Init();
    SystemClock_Config();
    MX_GPIO_Init();
    MX_ADCx_Init();
    MX_TIMx_Init();
    while (1) {
        read_command();
        validate_command();
        update_setpoint();
        check_timeout();
    }
}
```

ISR (ADC / TIM):

```
void ADC_IRQHandler() {
    // 1. baca arus fase
    ia = ADC1->DR;
    ib = ADC2->DR;
    ic = ADC3->DR;
    // 2. Clarke
      ialpha = ia,
      ibeta = (ia + 2*ib)/sqrt(3);
    // 3. Park
    id = ialpha*cos(theta) + ibeta*sin(theta);
    iq = -ialpha*sin(theta) + ibeta*cos(theta);
    // 4. PI
    vd = pid_d(id_ref - id);
    vq = pid_q(iq_ref - iq);
    // 5. Inverse Park + SVPWM
    set_pwm_compare(vd, vq);
    // 6. cek fault
      check_fault);
}
```

Nama simbol, register, dan konfigurasi interrupt harus dicocokkan dengan source code serta board pada branch yang digunakan.

VII. VARIABEL GLOBAL

(contoh, representatif)

```
volatile uint8_t motor_state;
volatile uint8_t fault_code;
volatile uint8_t uart_rx_ready;
volatile float target_speed;
volatile float target_current;
volatile float phase_current_a;
volatile float phase_current_b;
volatile float phase_current_c;
volatile float id_ref;
volatile float iq_ref;
volatile float id_meas;
volatile float iq_meas;
volatile float electrical_angle;
volatile float rotor_speed;
```

VIII. PRIORITAS INTERRUPT

No	Handler	Prioritas	Keterangan
1	ADC / Control Timer	0 (tertinggi)	Sampling & kontrol arus
2	PWM Update / Timer	1	Update duty cycle
3	UART Receive	2	Terima data command
4	Hall / EXTI	3	Input rotor/Hall
5	SysTick	4 (terendah)	Task periodik

Catatan: Nilai prioritas NVIC harus disesuaikan dengan konfigurasi pada firmware yang digunakan.

IX. RUMUS TRANSFORMASI & KONTROL (FOC)

Clarke Transformasi ($3 \rightarrow 2$)

$$i_\alpha = i_a$$
$$i\beta = (i_a + 2ib)/\sqrt{3}$$
$$(i_a + i_b + i_c = 0)$$

Park Transformasi ($\alpha\beta \rightarrow dq$)

$$i_d = i_\alpha \cos\theta + i\beta \sin\theta$$
$$i_q = -i_\alpha \sin\theta + i\beta \cos\theta$$

PI Controller

$$u(t) = Kp \cdot e(t) + \underline{Ki} \int e(t)\,dt$$

Keterangan: Rumus di atas merupakan referensi umum pada kontrol FOC, bukan jaminan bahwa rumus yang sama persis digunakan di dalam source code.